26900197 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEECT309 - ANALOG AND DIGITAL COMMUNICATION

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which statement best compares analog and digital communication? [U] / [CO1]

   a). Both systems always use identical signal forms

   b). Analog communication uses only binary signals

   c). Analog communication uses continuously varying signals, whereas digital communication uses discrete signal levels

   d). Digital communication cannot be affected by noise

2. When the frequency of the modulating signal is increased while its amplitude remains constant, the FM modulation index: [E] / [CO2]

   a). Increases b). Decreases

   c). Remains infinite d). Becomes zero

3. What is baseband communication? [K] / [CO3]

a). Transmission using a carrier signal

b). Transmission of the original information signal without modulation

c). Transmission through satellite only

d). Transmission using microwave signals

4. Which components form the voltage divider in a Hartley oscillator? [K] / [CO4]

a). C_1 and C_2

b). R_1 and R_2

c). L_1 and L_2

d). R and C

5. Why is a high-frequency carrier signal used for modulation? [U] / [CO1]

a). To increase noise

b). To facilitate efficient transmission and reduce antenna size

c). To eliminate the message signal

d). To decrease the carrier frequency

6. The staircase approximation in delta modulation is generated using a: [U] / [CO3]

a). Differentiator b). Integrator c). Oscillator d). Mixer

7. The bandwidth of a conventional AM wave is: [U] / [CO2]

a). 2fm

b). fm

c). 2fc

d). fc+fm

8. DPCM is most suitable for signals where: [U] / [CO3]

a). Adjacent samples have strong correlation

b). Adjacent samples are completely unrelated

c). There is no redundancy

d). Sampling is impossible

9. In AM, which property of the carrier signal is varied? [K] / [CO4]

a). Frequency

b). Amplitude

c). Phase

d). Speed

10. Which of the following is an FM detection method? [K] / [CO2]

a). Peak detector

b). Envelope detector

c). Product detector only

d). Slope detector

11. The bandwidth of a signal is defined as the difference between its: [U] / [CO1]

a). Maximum amplitude and minimum amplitude

b). Highest component frequency and lowest component frequency

c). Transmission distance and wavelength

d). Input voltage and output voltage

12. In the filter method of SSB generation, an AM/DSBSC signal with carrier frequency fc and message frequency fm is passed through a [U] / [CO2]

a). Low-pass filter

b). High-pass filter

c). Notch filter

d). Band-pass filter

13. Granular noise occurs when: [U] / [CO3]

a). Step size is too small for a rapidly changing signal

b). Step size is too large compared with slowly varying input

c). Sampling frequency is zero

d). There is no quantizer

14. In binary FSK, how many different frequencies are used? [U] / [CO4]

a). One

b). Two

c). Three

d). Four

15. In a PCM transmitter block diagram, which component comes immediately after the analog low-pass filter? [K] / [CO3]

a). Encoder

b). Sampler

c). Quantizer

d). Low pass filter

16. The frequency of a Colpitts oscillator mainly depends on which components? [U] / [CO4]

a). Resistors only

b). Transistor only

c). Inductor and capacitors

d). Diode and resistor

[1x16 = 16]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

17. What is modulation?Classify and Explain different types of modulation [K] / [CO1]

18. Derive the mathematical expression for a PM wave and explain phase deviation, modulation index, and instantaneous frequency. [A] / [CO2]

19. Explain Binary Frequency Shift Keying (BFSK) with suitable waveforms. [A] / [CO3]

20. Explain the generation of PSK with diagram and waveform. [U] / [CO4]

21. List any four advantages of modulation in a communication system. [K] / [CO1]

22. Explain the direct method of FM generation. [U] / [CO2]

23. Draw and explain the block diagram of an ASK transmitter and receiver. [U] / [CO3]

24. Explain the principle and generation of Binary ASK (BASK). [U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **12** marks

25. Explain the signal flow in a communication system from the information source to the destination with the help of a block diagram. [U] / [CO1]

26. Derive the mathematical expression for a DSBSC signal. Explain the generation of DSBSC using a balanced modulator with a neat block diagram. [An] / [CO2]

27. What is companding? Explain the process of compression and expansion in a PCM system. Discuss the need, advantages, and applications of companding. [U] / [CO3]

28. Design a High Pass Filter with a cut-off frequency of 1 kHz and explain its frequency response. [C] / [CO4]

[12x2 = 24]